

Jonathan Bourget

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Profile

Electrical Engineering student with hands-on expertise in integrated hardware-firmware systems. Experienced in architecting embedded platforms with wireless communications, power delivery, and real-time control. Proven ability to identify technical constraints, design solutions, and validate performance in real-world conditions.

Education

BENG Bachelor of Electrical Engineering, Co-op Option — Carleton University Sept 2024 – Present

- 3rd Year Standing, 9.97/12 GPA (B+) | Carleton Academic Scholarship Recipient
- **Expected Graduation:** April 2029

Hardware & Systems Projects

Avionics & Power Distribution PCB — RC Drone Platform *Team Lead, Ongoing*

- Led electronics subsystem for multi-rotor drone: designed multi-rail power distribution board (raw battery → 12V, 5V, 3.3V regulated outputs) in Altium Designer with thermal analysis. Implemented STM32 flight controller with BNO085 9-axis IMU and ExpressLRS wireless protocol for real-time vehicle control.
- **Skills:** PCB design (multi-layer), embedded C, wireless communications (RC protocols), power electronics, systems integration

Electric Go-Kart — Full Cycle Design & Optimization *Personal Project, 2019*

- Designed and built 1.6kW electric drivetrain achieving 49 km/h top speed and 5+ hour runtime. Hand-welded steel frame, integrated 4×12V lead-acid battery pack, designed custom motor controller, and optimized power delivery through iterative testing.
- **Skills:** Power systems design, motor control, thermal management, mechanical fabrication, requirements-driven iteration

Full-Stack AI Secretary — Embedded Systems Integration *Personal Project, Ongoing*

- Architected and deployed self-hosted AI agent on Orange Pi Zero 3 (ARM Linux). Backend integrates Gmail, Google Calendar, and custom data pipelines via APIs; frontend provides conversational interface. Designed with modular architecture to support expanding capabilities—currently exploring AI applications in personal finance and data analysis.
- **Skills:** Embedded Linux, full-stack development, API integration, system architecture, database design

Modular Bicycle Lighting System — Custom PCB Design *Personal Project, 2022*

- Designed weather-resistant RGBW lighting system with custom Arduino-controlled PCB and 3D-printed enclosures.
- **Skills:** PCB design (EasyEDA), embedded C/Arduino, firmware, CAD modeling, product design

Academic & Coursework Projects

Bicycle Network Infrastructure Proposal — RFID Feasibility Study *Team Project, 2026*

- Conducted systems analysis for RFID-based smart bicycle infrastructure at Carleton University. Researched RF technologies, evaluated bandwidth requirements and existing systems, assessed user needs against feasibility constraints, and synthesized findings into formal proposal and executive presentation.
- **Skills:** Systems analysis, technology research, requirements definition, technical writing, team collaboration

Work Experience

Lifeguard, Swim Instructor & First Aid Instructor — City of Ottawa

- Safety-critical role requiring rigorous documentation, protocol compliance, and rapid decision-making in high-liability environments. Assess dynamic situations, determine response procedures, coordinate team actions under time pressure.
- Certified: Standard First Aid, CPR-C with AED, Airway Management & Oxygen Administration, SFA Instructor/Examiner

Technical Proficiencies

PCB & Circuit Design: Altium Designer, KiCad, EasyEDA, schematic capture, multi-layer PCB layout, power delivery design, thermal analysis

Embedded Systems & Microcontrollers: Arduino, STM32, ESP32, Orange Pi (Linux/ARM), real-time firmware

Programming: C/C++ (embedded focus), Python, Verilog HDL, Java

Wireless & Communications: ExpressLRS, RFID, wireless systems analysis, signal integrity considerations

Design, Simulation & Test: Autodesk Fusion CAD, FPGA design (Vivado), oscilloscope/multimeter, breadboarding, hardware validation

Other: Git version control, LaTeX, technical writing, process documentation, requirements analysis